

Md Abu Sayed, Ph.D.

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SUMMARY

Applied AI / Machine Learning Scientist with a Ph.D. in Computer Science and Engineering and expertise in deep learning, temporal sequence modeling, medical computer vision, and generative AI. Proven track record architecting production AI systems (LLM/RAG, vector search, recommendation engines) and state-of-the-art predictive frameworks (Transformers, GANs, multi-class segmentation) for safety-critical healthcare and autonomous systems. 12 peer-reviewed publications (7 first-author; 2 journals).

TECHNICAL SKILLS

Machine Learning & CV:	PyTorch, TensorFlow, Keras, OpenCV; multi-class segmentation, Transformers, sequence models (LSTM, GRU), generative models (GAN, WGAN-GP, CVAE), multimodal fusion, explainable AI (XAI)
Agentic AI & Systems:	LangChain, agentic workflow orchestration, RAG / multimodal RAG, vector search (pgvector, HNSW), FastAPI, PostgreSQL, REST APIs, Git, Docker
Languages & Simulation:	Python, C/C++, C#, SQL, MATLAB Unity 3D engine, custom HLSL/Cg shaders, TCP socket IPC

EXPERIENCE

Postdoctoral Researcher, AI Initiatives Jun 2026 – Present

Center for Digital Cardiovascular Innovations, University of Miami Miller School of Medicine *Miami, FL*

- Developing deep neural network pipelines for multi-class segmentation of lumen, vessel wall, and plaque composition (calcium, fibrous, fibrolipidic) across **HD-IVUS** and **NIRS-IVUS** pullbacks, trained on **40,000+ expert-labeled frames** and benchmarked against expert inter-observer variability (*Scientific Reports*, under review).
- Automated IVUS-based calcium morphology quantification to forecast stent under-expansion prior to intervention.
- Architecting clinical decision-support models for PCI lesion-preparation strategy selection (lithotripsy, atherectomy, balloons) trained on a virtual simulation corpus spanning **1,200 lesion phenotypes and patient-specific anatomies**.

AI Software Engineer Intern

Aug 2025 – May 2026

Pluto in Aquarius LLC

Reno, NV

- Architected and deployed **ICONIQ**, an end-to-end multimodal recommendation service combining dense embedding generation, pgvector approximate nearest neighbor (ANN) retrieval, and low-latency re-ranking on FastAPI / PostgreSQL.
- Designed the backend for **AIJINTSEE**, an agentic marketing-automation platform featuring LangChain orchestration, retrieval-augmented generation (RAG), and multi-provider LLM routing with structured output validation.
- Led and mentored a 4-engineer team (2 M.S., 2 B.S.) on system architecture, coding standards, and rigorous experiment design.

Graduate Research Assistant (ONR-Funded)

Aug 2021 – May 2026

Department of Computer Science and Engineering, University of Nevada, Reno

Reno, NV

- Architected **MTITP**, a multi-task Transformer with a 3-stream encoder (kinematics, pairwise geometry, symbolic COLREGs) that jointly infers vessel intent and predicts trajectories: **99.2% intent accuracy**, **99.4% future intent mAP**, and **2.3× lower displacement error** than Kalman filter baselines across 880K+ encounter windows.
- Developed **MarITGAN**, an intent-aware GAN critic that cut trajectory error by **35%** over generic WGAN-GP critics while preserving multimodal sampling diversity; demonstrated that intent conditioning directly improves trajectory prediction.
- Lead developer of **NavySim**, a high-performance Unity multi-vessel simulator with a real-time TCP socket bridge for low-latency deep learning inference, engineered for on-water deployment by **Huntington Ingalls Industries** (*IEEE Trans. on Games* 2026).
- Developed **ThreatMap**, fusing sensor coverage, kinematic vulnerability fields, and CPA metrics into dynamic threat heatmaps.

SELECTED PUBLICATIONS

- **M. A. Sayed** et al. NavySim 2.0: Enhanced multi-vessel simulation and analysis engine for advanced naval research. *IEEE Transactions on Games*, 2026.
- **M. A. Sayed** et al. Early classification of intentions for maritime domains using deep learning models. *IEEE CASE*, 2025.
- W. Wu, ..., **M. A. Sayed**, ..., Y.S. Chatzizisis. Deep learning model for multi-class segmentation of coronary high-definition IVUS. *Under review, Scientific Reports*, 2026.
- **M. A. Sayed** et al. An innovate approach for retinal blood vessel segmentation using mixture of supervised and unsupervised methods. *IET Image Processing*, 2021.

EDUCATION

Ph.D., Computer Science and Engineering, University of Nevada, Reno

Aug 2021 – May 2026

Dissertation: *Deep Generative and Explainable Learning Frameworks for Intent Recognition in Naval Domain*

M.S., Computer Science and Engineering, University of Nevada, Reno (earned en route to Ph.D.)

Aug 2021 – Dec 2023

B.S., Computer Science and Engineering, Khulna University, Bangladesh

Dec 2014 – Jan 2019